Architectural description of the system

• Detailed System Description

The system is designed as a role-playing game (RPG) where

players navigate through a virtual world, interact with

various elements, and engage in combat and quests. The

game is structured to provide a rich and interactive

experience.

Genre: Interactive fiction

• Requirements and Usage Scenarios

1. Player Management:

- Attributes: Players have health points (HP), stamina, and

other attributes that affect gameplay.

- Actions: Players can perform actions such as moving, attacking, using items, and interacting with the

environment.

- Inventory: Players can manage their inventory by

adding, removing, and using items. They can also equip and

unequip weapons and shields.

2. Combat System:

- Engagement: Players can initiate combat with enemies

present in the game world.

- Mechanics: Combat involves calculating damage based

on player and enemy attributes, weapons, and shields.

- Outcomes: Combat results in changes to player and

enemy health, potentially leading to victory or defeat.

3. Quest System:

- Objectives: Quests have specific objectives that players
must complete to progress.

- Progression: Players can start, progress, and complete

quests, with progress tracked and rewards given upon

completion.

4. Environment Interaction:

- Locations: Players can move between different

locations, each with its own set of interactive elements.

- Interactive Elements: Players can interact with things

like doors, chests, and beds, which may reveal items or

trigger events.

• **Technical Requirements for the Game**

1.Platform Compatibility

- Operating System: The game must run on Ubuntu 20.04
LTS or later.

2. System Requirements

- C++ (google unit testing framework, version C++11

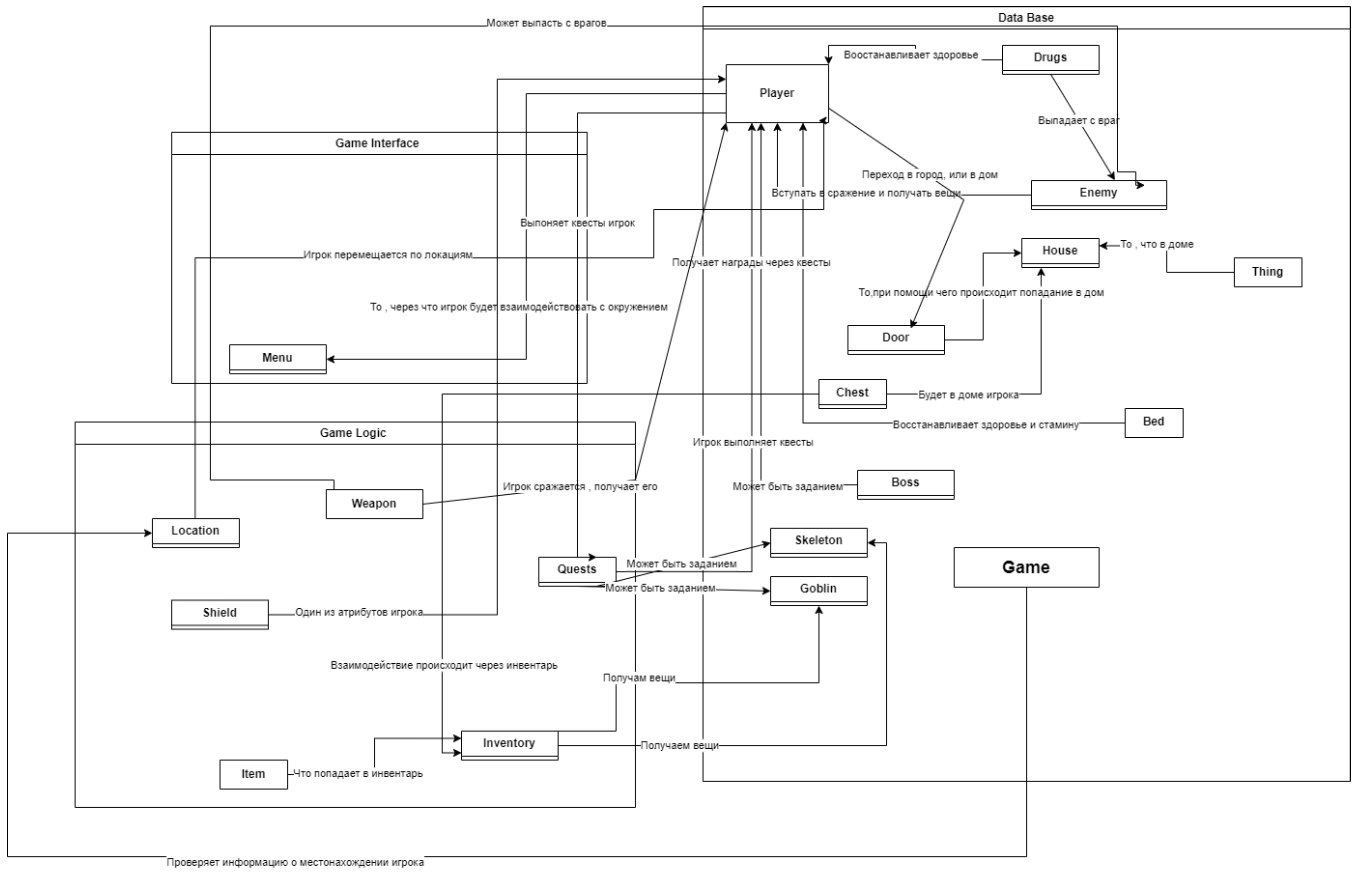
and later).

This list covers the technical requirements for the game,

including its compatibility with Ubuntu, system

requirements, and necessary software dependencies.

- **Component Diagram Description**

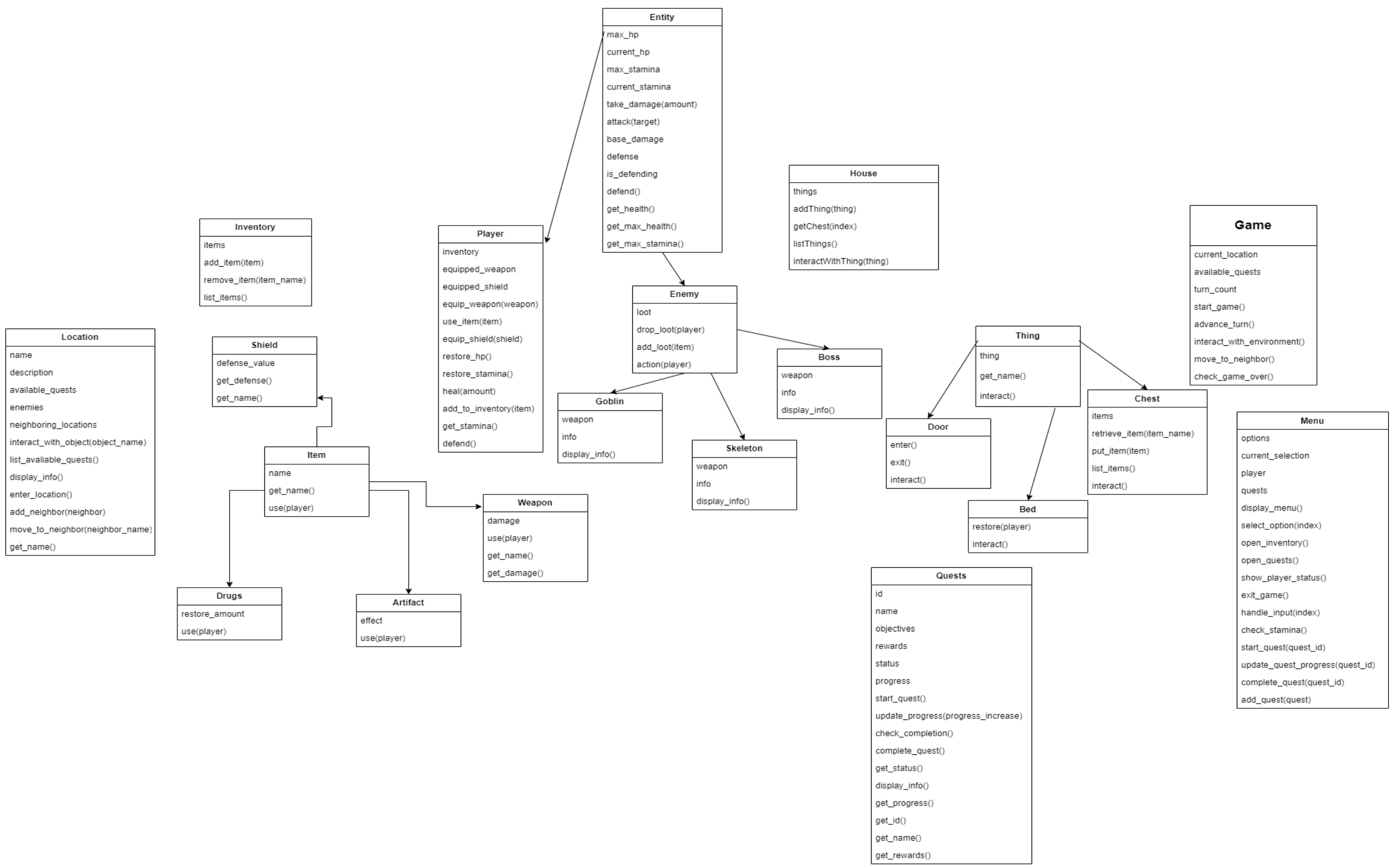


The component diagram provides a high-level view of the

system architecture, highlighting the main components and

their interactions.

- **Class Diagram Description**



The class diagram offers a detailed look at the system's classes, their attributes, methods, and relationships.

- **Detailed Test Plan**

1. Unit Testing:

- Classes and Methods: Test each class and its methods

individually to ensure they function correctly.

- Coverage: Include tests for inventory management,

combat calculations, quest progression, and environment

interactions.

Results of the test are provided in the same folder.